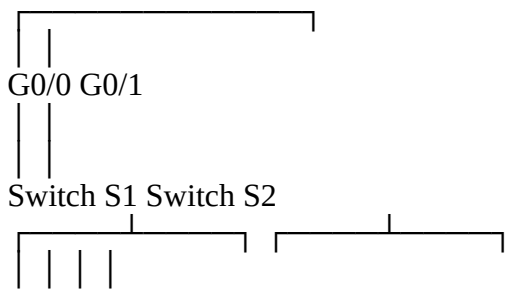
Ex.No.10 Design a simple topology and configure with one router, two

switches and PCs using Cisco Packet Tracer

DATE:
AIM:
To design and configure a simple network topology with one router, two switches, and four PCs in Cisco Packet Tracer for successful inter-network communication.
PROCEDURE:

1. Topology
Router R1



2. Devices to place
In Cisco Packet Tracer, place:
- 1 Router – Cisco 1941
- 2 Switches – Cisco 2960
- 4 PCs

Use Copper Straight-Through cables.
3. IP Addressing

Device	Interface	IP Address	Subnet Mask	Gateway
R1	G0/0	192.168.1.1	255.255.255.0	—
R1	G0/1	192.168.2.1	255.255.255.0	—
PC1	NIC	192.168.1.10	255.255.255.0	192.168.1.1
PC2	NIC	192.168.1.11	255.255.255.0	192.168.1.1
PC3	NIC	192.168.2.10	255.255.255.0	192.168.2.1
PC4	NIC	192.168.2.11	255.255.255.0	192.168.2.1

So we have two different networks:

Network 1: 192.168.1.0/24
PC1 ─┐
PC2 ─┤
S1 ─── R1 G0/0 = 192.168.1.1
Network 2: 192.168.2.0/24
PC3 ─┐
PC4 ─┤
S2 ─── R1 G0/1 = 192.168.2.1
4. Configure Router R1
Click R1 → CLI and enter:

```
enable
configure terminal
interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
```

exit
interface gigabitEthernet 0/1
ip address 192.168.2.1 255.255.255.0
no shutdown
exit
end
write memory
The important command is:
no shutdown
It activates the router interface.
Check the interfaces using:
show ip interface brief
You should see the interfaces as up/up.
SCREENSHOTS:(to be pasted)

5. Configure PC1
Go to:
PC1 → Desktop → IP Configuration
Enter:
IP Address: 192.168.1.10
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1
SCREENSHOTS:(to be pasted)

6. Configure PC2
IP Address: 192.168.1.11
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1
SCREENSHOTS:(to be pasted)

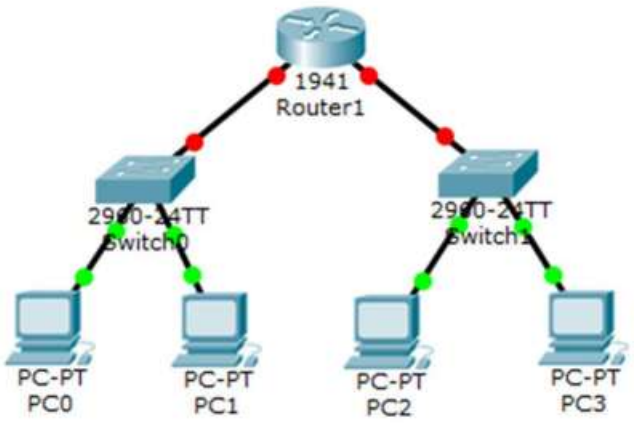
7. Configure PC3
IP Address: 192.168.2.10
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.2.1
SCREENSHOTS:(to be pasted)

8. Configure PC4
IP Address: 192.168.2.11
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.2.1
SCREENSHOTS:(to be pasted)

9. Test the network
From PC1 → Desktop → Command Prompt, type:
ping 192.168.1.11
This tests PC1 → PC2.
Then:
ping 192.168.2.10
This tests PC1 → Router → Switch 2 → PC3.
You should receive:
Reply from 192.168.2.10
Reply from 192.168.2.10

Reply from 192.168.2.10
Reply from 192.168.2.10
SCREENSHOTS:(to be pasted)

Sample Configuration:



Configure Terminal:

```
If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISCO1941/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
2 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)


         --- System Configuration Dialog ---

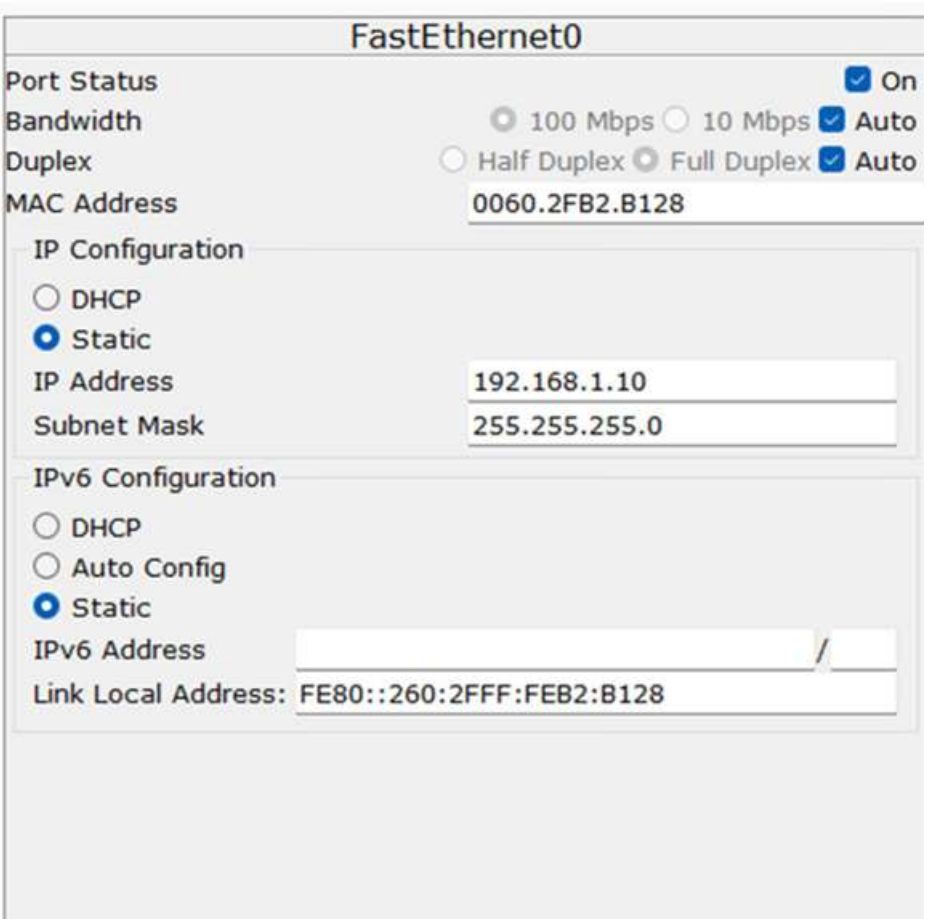
Continue with configuration dialog? [yes/no]: n


Press RETURN to get started!



Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#no ip address
```

IP Addressing:



Result:

Thus the experiment to design simple topology and configure with the one router, two switches and PCs using Cisco Packet Tracer has been done and verified the output.